How-To Library

Informatica

Informatica Enterprise Data Catalog 10.5 Security Guide

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Abstract

This article provides information about the various security features available in Enterprise Data Catalog. Use the Informatica Enterprise Data Catalog Security Guide to learn about how to implement and manage the security components in your Enterprise Data Catalog installation. Understand the functioning of the authentication process and learn how to manage users and groups, and configure permissions, privileges, and roles.

Supported Versions

- Informatica Enterprise Data Catalog 10.5

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Overview

You can secure the Enterprise Data Catalog installation to protect the catalog from threats inside and outside the network. Security for the Informatica domain includes both infrastructure security and operational security. Use the Enterprise Data Catalog Security Guide to learn about how to use the infrastructure security features available in the Informatica domain. Additionally, understand how to configure and manage the authentication protocols. You can enable secure communication to the Informatica domain using SSL certificates.

The Infrastructure security protects the Informatica domain from unauthorized access or modification of services and resources in the Informatica domain.

Infrastructure security includes the following aspects:

- Protection of data transmitted and stored within the Informatica domain
- Authentication of users and services connecting to the Informatica domain
- Secure connections for external components, including client applications and relational databases for repositories, sources, and targets

The operational security features help to manage users and groups, and set permissions, privileges, and roles. The operational security controls access to the data and services.

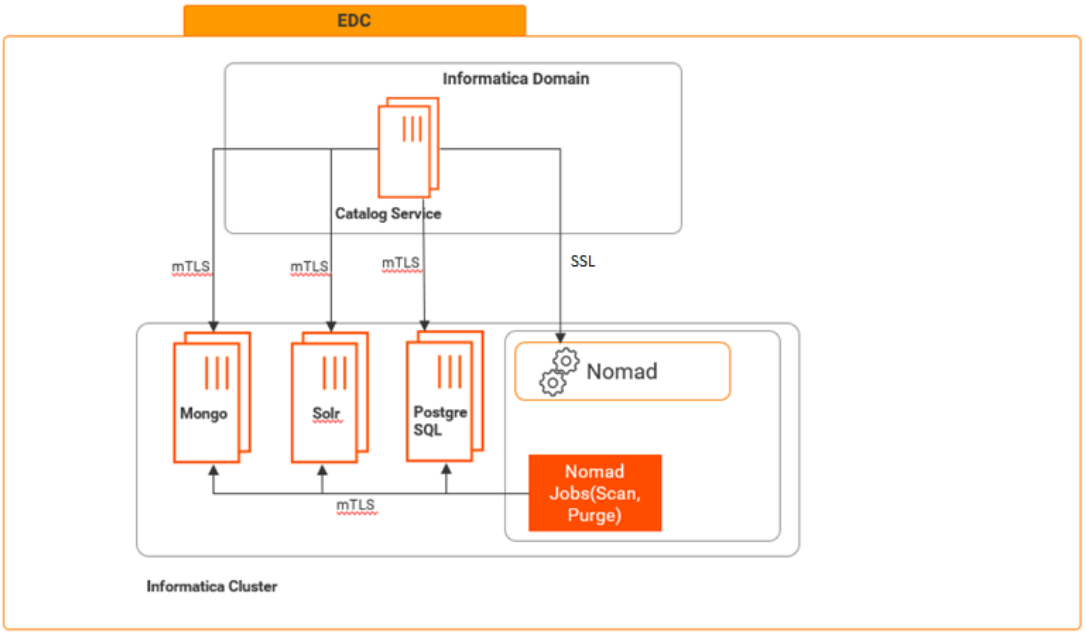
Operational security includes the following aspects:

- Role-based access permissions to data and metadata for users
- Role-based access permissions to perform operations

Informatica stores the domain configuration information and the list of users authorized to access the domain in the domain configuration repository. The domain configuration repository also contains the groups, roles, privileges, and permissions that are assigned to each user in the Informatica domain.

Authentication

The services running in the Informatica cluster use the Mutual Transport Layer Security (mTLS) protocol for communication over encrypted channels. The following diagram shows the communication between the services running in the cluster and the Catalog Service running in the Informatica domain.



SSH Authentication

The Secure Shell protocol also called as SSH is a cryptographic protocol that allows secure remote connections between two hosts. Using SSH, you can administer and manage remote hosts and also copy or move files through an encrypted channel. To log into a remote system over SSH, you can use two methods — password based authentication and public key-based authentication also called as key-based authentication. Key-based authentication allows passwordless login and is useful for running automation processes securely over a network.

You can use key-based authentication to perform installation tasks. Additionally, key-based authentication is used to perform system health monitoring.

During installation or upgrade, binary files, configuration files, certificates, and other installation files are copied from the Informatica domain to the gateway nodes, and then to the worker nodes. Additionally, when the Informatica Cluster Service performs the system monitoring and health checks for the associated services such as MongoDB, Nomad, Solr, Apache ZooKeeper, and other services, certain configuration files are copied from the domain server to the cluster nodes. You must enable key-based authentication to ensure that these operations are carried out seamlessly without manual intervention.

Enabling key-based authentication

You must enable key-based SSH authentication for the user that runs the services on the gateway and cluster nodes.

To enable key-based authentication, perform the following steps:

1. Generate a public key using the `ssh-keygen -t rsa` command.
2. Copy the key to the nodes using the `ssh-copy-id -i ~/.ssh/id_rsa.pub <username>@<hostname>` command.

The command prompts for the password for the remote user. After you authenticate using the password, the key is added to the `~/.ssh/authorized_keys` file in the home directory of the remote user.

User Authentication

Users can log in to Enterprise Data Catalog using the following types of authentication:

- **Native Authentication.** Native authentication mode is available only for user accounts in the Informatica domain. When the Informatica domain uses native authentication, the user credentials and privileges are stored in the domain configuration repository and is used to perform all user authentication within the Informatica domain. If the Informatica domain uses native authentication, by default, the domain has a Native security domain and all user accounts belong to the Native security domain. Informatica uses user names and passwords to authenticate users and services in the Informatica domain.
- **LDAP Authentication.** LDAP is a protocol to authenticate users and resources on a network. If the Informatica domain uses LDAP authentication, the user accounts and credentials are stored in the LDAP directory service. The user privileges and permissions are stored in the domain configuration repository. You must periodically synchronize the user accounts in the domain configuration repository with the user accounts in the LDAP directory service. Informatica uses user names and passwords to authenticate users and services in the Informatica domain.
- **SAML-based Single Sign-on.** Security Assertion Markup Language (SAML) is an XML-based data format for exchanging authentication and authorization information between a service provider and an identity provider. You can configure SAML-based single sign-on for Enterprise Data Catalog using the following identity providers:
  - Citrix NetScaler
  - Oracle Access Manager
  - Okta
  - PingFederate
  - F5 Big-IP
  - Azure Active Directory

Certificate-Based Authentication

To enable SSL certificate-based authentication for the Informatica domain, the following options are available:

- You can configure the Informatica Cluster Service to use the out-of-the-box (OOTB) certificates for mTLS authentication.
- You can configure custom SSL certificates for the Informatica domain.

If you choose the default option, Informatica Cluster Service uses the certificate chain stored in the keystore configured for the Informatica domain to generate the required cluster certificates.

If you have a custom SSL certificate for the Informatica domain that has a signing capability to sign other certificates, the Catalog Service and the Informatica Cluster Service generate all the required certificates.

If you have a custom SSL certificate for the Informatica domain, but the certificate does not have the capability to sign other certificates, you must use CA-signed certificates for the cluster and clients. The certificates are stored in a custom location on the domain host. The client certificates are used by the Catalog Service to authenticate to the services running on the Informatica cluster. The cluster certificates are used by Nomad to communicate with the associated services such as MongoDB, Solr, and PostgreSQL.

You can use the `generate_csr.sh` script bundled with the installer to generate the Certificate Signing Request (CSR) to submit to the external CA. You can then create the required custom SSL certificates using the `generate_certs.sh` script. To know more about how to generate the CSR and create the custom SSL certificates, see scenario 2 under Appendix C of the *Installation for Enterprise Data Catalog* guide.

The following table contains the details about the certificates and keys used for authentication and secure communication:

Certificate	File Name	Default File Path	Description
Domain keystore	infa_keystore.jks	`services/shared/security`	Used for service communication.
Domain truststore	infa_truststore.jks	`services/shared/security`	Used to access the Catalog Service and the Informatica Cluster Service REST APIs and all trusted certificates. The domain truststore is the same as the catalog truststore and the Informatica Cluster Service truststore.
Default keystore	Default.keystore	`tomcat/conf`	Used to establish secure connection to the Administrator tool.
Client keystore	keystore.jks infa_privkey.pem	`services/shared/security/$ICS/client_certs`	Used by the Catalog Service to communicate with other mTLS-enabled cluster services.
Client truststore	truststore.jks	`services/shared/security/$ICS/client_certs`	Used by the Catalog Service to communicate with other mTLS-enabled cluster services.
Client truststore certificates	infa_pubcert.pem infa_nodecert.pem infa_nodecertkey.pem	`services/shared/security/$ICS/client_certs`	Used by the Catalog Service to communicate with other mTLS-enabled cluster services.
Cluster keystore	keystore.jks infa_privkey.pem	`$customDir/certs` on the cluster node	Used by the Informatica Cluster Service to launch other mTLS-enabled services.
Cluster truststore	truststore.jks	`$customDir/certs` on the cluster node	Used by the Informatica Cluster Service to launch other mTLS-enabled services.
Cluster truststore certificates	infa_pubcert.pem infa_nodecert.pem Infa_nodecertkey.pem	`$customDir/certs` on the cluster node	Used by the Informatica Cluster Service to launch other mTLS-enabled services.

You can use the `GenerateCustomSslUtility` utility bundled with the installer to generate the required custom SSL certificates that have the capability to sign other certificates. You can sign the certificates using a CA bundled within the utility. To generate the required custom SSL certificates using the `GenerateCustomSslUtility` utility, perform the following steps:

1. Extract `GenerateCustomSslUtility.zip` from the following location: `<Location of installer files>/properties/utils/CustomSslCertsUtility/`.
2. Set the JAVA_HOME environment variable to point to JDK 8.

3. Configure the following parameters in the `input.properties` file that you extracted from the `GenerateCustomSslUtility.zip` file:

Parameter	Description
KeystoreFile	Path to a custom keystore file along with file name. The keystore type must be in JKS. X509 format. The file must contain a single PrivateKeyEntry with the complete certificate chain. Verify that the file has CA-signing capability.
KeystorePassword	Password of the custom KeystoreFile in plain text format.
TruststoreFile	Path to a custom truststore file along with file name. the truststore type must be in JKS. X509 format. The file must contain the public certificates corresponding to the PrivateKeyEntry in the KeystoreFile.
TruststorePassword	Password of the custom TruststoreFile in plain text format.
ISPDomainKeystorePassword	Password of Informatica domain keystore in plain text format. The utility uses the password for the generated Informatica Cluster Service cluster and client keystore.jks file.
ISPDomainTruststorePassword	Password of the Informatica domain truststore in plain text format. The utility uses the password for the generated Informatica Cluster Service cluster and client truststore.jks file
KeystoreOutputDir	Represents the location of the Informatica domain custom keystore that you provided when you installed Enterprise Data Catalog.
TruststoreOutputDir	Represents the location of the Informatica domain custom truststore that you provided when you installed Enterprise Data Catalog. **Note:** Verify that the KeystoreOutputDir and TruststoreOutputDir parameters point to the same directory.
ClusterNodes	Comma-separated list of fully qualified host names of nodes that you plan to configure as data nodes, processing nodes, service hosts, and gateway node when you configure the Informatica Cluster Service.
IcsServiceName	Name of the Informatica Cluster Service.
DomainNodes	Comma-separated list of fully qualified domain names of nodes that you plan to configure as Informatica domain hosts.
IspDomainName	The Informatica domain name.
ClusterNodeDNSDomain	Domain name of the gateway host that you plan to configure for the Informatica Cluster Service.

4. Run the utility using the following command: `java -jar GenerateCustomSslUtility.jar -in input.properties`. The utility generates the following keys and client and cluster certificates:
   - Keys:
     - keystore.jks

- infa_privkey.pem

  Note: The cluster keys are generated in the following directory: `<KeystoreOutputDir>/<IcsServiceName>/cluster_certs`. The client keys are generated in the following directory: `<KeystoreOutputDir>/<IcsServiceName>/client_certs`

- Certificates:

  - truststore.jks
  - infa_pubcert.pem
  - infa_nodecert.pem
  - infa_nodecertkey.pem

  Note: The cluster certificates are generated in the following directory: `<TruststoreOutputDir>/<IcsServiceName>/cluster_certs`. The client certificates are generated in the following directory: `<TruststoreOutputDir>/<IcsServiceName>/client_certs`

5. Assign the ownership of the following directories to the Informatica domain user:
   - <KeystoreOutputDir>/<IcsServiceName>/cluster_certs
   - <TruststoreOutputDir>/<IcsServiceName>/cluster_certs
   - <KeystoreOutputDir>/<IcsServiceName>/client_certs
   - <TruststoreOutputDir>/<IcsServiceName/client_certs

Note: Verify that all the client certificates are included in a single directory. Similarly, verify that all the cluster certificates are included in a single directory. You must also verify that the directories have the `chmod 700` permission configured and the files under the directories have the `chmod 600` permission configured.

Authorization

Different privileges and permissions are provided by the administrators to the Enterprise Data Catalog users to control their access to the catalog. Privileges determine the actions that users can perform in the catalog. Privileges indicate permissions to perform an action in Enterprise Data Catalog. You can associate privileges with users and groups using the Administrator tool. A role is a collection of privileges that you can assign to users and groups. The administrator assigns roles or privileges to users and groups for the domain and for the application services in the domain. To access the catalog, a user must have the catalog privileges.

As an Informatica Administrator, you can grant the following privileges to the catalog users:

- **Catalog.** Users and groups with the catalog privilege can view the asset metadata, and other data including data based on data domain discovery. The users can also edit the metadata and perform data curation.
- **Administration.** Users and groups with the administration privilege can perform administrative tasks such as resource management, domain and attributes management, and service and task monitoring.
- **Development.** Users and groups with the development privilege has full access to perform any actions including access to all REST APIs.

The permissions that you grant to users can be categorized into two types:

- **Config permissions.** The config permissions are a set of permissions that you can use to control the actions a user can perform with a resource or scanner. For example, you can assign the permissions for users to perform a scan, perform connection assignments, modify security, or edit resource configuration.

- **Data permissions.** You can use data permissions to control the level of access that users or groups have on the objects extracted by a specific resource. Based on the data permissions, users who log into the catalog can view the resources and perform various operations on the resources. For example, you can assign the read permission to allow users to view objects, and assign the write permission to allow users to perform enrichments on the objects. You can also assign the metadata read and data read permissions to allow users to view profiling information for the objects.

Supported User Personas

You can configure three personas as special attributes for a resource:

1. Data owner
2. Data steward
3. Subject matter expert

You can assign these personas to catalog users. The users can then enrich the catalog assets by various means. For example, a data owner or a data steward can certify trustworthy data assets based on their analysis. The certified data assets can be associated with comments or searchable tags. Certified data assets are ranked higher in the search results.

If a user is associated to a data resource as one of the three personas, the user gets the read and write data permissions over the resource irrespective of the previous permissions that the user was associated with the resource.

Encryption

Informatica encrypts sensitive data, such as passwords and secure connection parameters before it stores the data in the domain configuration repository or the Model Repository Service database. Informatica also saves sensitive files, such as configuration files, in a secure directory. Sensitive information is encrypted using the AES-256 algorithm.

The PostgreSQL database also stores sensitive information such as value frequencies in encrypted form.

Firewall Configuration

The default ports for Nomad, MongoDB, Solr, ZooKeeper, and PostgreSQL must be accessible through your firewall.

The details for the ports used by the associated services is as follows:

Service	Default Port	Description
Nomad	4646	Used by clients and servers to serve the HTTP API. TCP connections only.
	4647	Used for internal RPC communication between client agents and servers. Also used for inter server traffic. TCP connections only.
	4648	Used for inter server traffic. TCP and UDP connections.
MongoDB	27017	Used by the mongod service for communication.
Solr	8983	Used as the client port for the Solr service.
ZooKeeper	2181	Used as the client port for the ZooKeeper service.
	2888	Used as the peer port for followers to connect the leader.

Service	Default Port	Description
	3888	Used as the leader port for leader election.
PostgreSQL	5432	Used to connect to the PostgreSQL service.

User Privileges

When you install Enterprise Data Catalog or perform any other administrative tasks, you run commands as a non-root user with sudo privileges. Sudo privileges are required by the services associated with the Informatica Cluster Service such as Apache Solr, PostgreSQL, and Nomad.

A non-root user can install Enterprise Data Catalog in general mode or in sudo less mode. In general mode, the user has all the sudo privileges, whereas in sudo less mode, the user only has the minimum privileges that are required to enable or disable services.

To enable a user to perform the installation in sudo less mode, the required commands must be added to the `/etc/sudoers` file on all the cluster nodes as follows:

```
<username>=(ALL) NOPASSWD: /bin/mkdir, /bin/chown, /bin/chmod, /bin/tee, /bin/echo, /bin/systemctl, /bin/cp, /bin/mv, /bin/tar, /bin/find, /bin/unzip, /usr/sbin/sysctl, /bin/rm, /bin/umount
```

To enable a user to perform the installation in general mode, you must add the following line to the `/etc/sudoers` file on all the cluster nodes:

```
<username>=(ALL) NOPASSWD: ALL
```

Note: `/bin` is the default directory. Make sure that you specify the correct directory for all the default commands. You can use the `which` command to identify the directory for a command.

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